

CSCE 640 QUANTUM ALGORITHMS · MODULE 10 · LECTURE NOTES

Noise and Error Correction

Why this is hard, and why it is nonetheless possible.

MODULE OUTCOMES

- Describe the error model and why it is continuous.
- Explain how a quantum code corrects without measuring data.
- State the threshold theorem and what it establishes.
- State the overhead, numerically.
- Read a resource estimate critically.

1 The error model

ERRORS ARE CONTINUOUS, BUT CORRECTING THE PAULI BASIS CORRECTS EVERYTHING

A qubit can be rotated by any small angle, and can decohere partially — so the error set is a continuum, which appears to rule out the discrete correction that every classical code relies on, and was the reason many people initially believed quantum error correction impossible.

But any single-qubit error decomposes into a combination of **I, X, Y, and Z** — they span the space of single-qubit operators — and the syndrome measurement projects the error onto that basis, which *discretises* it: a small rotation becomes, with high probability, no error at all, and occasionally a full X or Z.

So correcting bit flips (X) and phase flips (Z) corrects arbitrary single-qubit errors — which is the central and genuinely surprising fact of this module, and the one that made the field possible.

And phase flips are the new problem — classical codes have **no analogue**, because classical bits have no phase — which is what the quantum codes had to invent, and why Shor's nine-qubit code and its successors look unlike classical repetition codes.

2 Correcting without looking

THE PROBLEM

measuring the data destroys the superposition (Module 02), and no-cloning forbids making copies to compare.

THE TRICK

measure PARITIES, not values.
"do qubits 1 and 2 agree?" is a question whose answer does not reveal what either one is.

SO

encode one logical qubit across many physical ones, in a subspace defined by a set of commuting parity operators (stabilisers)
measure those operators repeatedly
an error moves the state out of the subspace
and the parity pattern (the syndrome) says which error occurred, without ever naming the state

THE SURFACE CODE

stabilisers are local on a 2D grid, which is why hardware likes it. Distance d corrects $(d-1)/2$ errors and costs about d^2 physical qubits per logical one.

Measure parities, not values — a parity answer carries no information about the individual amplitudes, so the superposition survives the measurement. This is the single idea behind every **quantum code**, and it is worth checking by hand on the three-qubit bit-flip code: the parity of qubits 1 and 2, and of 2 and 3, together identify which qubit flipped, and neither parity says anything about whether the logical state was 0, 1, or a superposition. The surface code's locality is what makes it the hardware favourite: every stabiliser involves only neighbouring qubits on a grid.

3 The threshold theorem

BELOW A THRESHOLD PHYSICAL ERROR RATE, ARBITRARILY LONG COMPUTATIONS ARE POSSIBLE WITH POLYLOGARITHMIC OVERHEAD

Concatenate a code with itself and the logical error rate falls doubly exponentially in the number of levels, while the overhead grows only exponentially — so below some physical error rate, correction wins, and above it, each level of encoding makes things worse.

Which establishes that noise is not a fundamental obstacle — it is an engineering one — and before this result it was genuinely unclear which it was, with serious people arguing that quantum computing was impossible in principle.

The threshold is roughly 1% for the surface code under favourable assumptions about the noise model and the measurement speed — and current hardware is at or near it, which is the real experimental progress of the last decade and the thing worth tracking.

But 'above threshold' is the beginning rather than the end — the overhead at a physical rate only just below threshold is enormous, because the code distance required grows rapidly as the margin shrinks — and that is §4.

4 The overhead

- Roughly a thousand physical qubits per logical qubit, at plausible near-term error rates with the surface code — and it rises steeply as the physical rate approaches the threshold, which is why a factor-of-two improvement in physical error rate can be worth far more than a factor of two in qubit count.
- Plus magic state distillation for the T gates (Module 03 §1) — which frequently dominates the entire footprint of a fault-tolerant circuit, since T gates cannot be done transversally and each one consumes a distilled resource state — and is why T-count rather than gate count is the metric.
- And the syndrome must be measured continuously and decoded faster than errors accumulate — so classical decoding throughput is itself a hard real-time engineering requirement, and is an active research area in its own right.
- Which is where Module 08 §4's millions of physical qubits come from: a few thousand logical qubits, times about a thousand, plus the distillation factories, which in some designs occupy most of the chip.
- So every resource estimate is really an estimate about error correction — and reading one means reading its assumed physical error rate first, because that assumption drives the overhead non-linearly and is the parameter that varies most between papers.

AND THIS IS WHERE THE HONEST OPTIMISM LIVES

The threshold theorem is a real result and it is good news — it says the problem is solvable in principle, which was genuinely open before it was proved and is the single most important theorem for the field's existence.

And the experimental progress is real: physical error rates have fallen steadily over two decades, and logical qubits that outperform their physical constituents have now been demonstrated — which is the milestone that actually mattered, and it was not obvious it would arrive when it did.

While the gap to a useful machine remains three to four orders of magnitude in qubit count (Module 08 §4) — **and both of those statements are true at the same time**, which is the whole difficulty of describing this field accurately.

Which is the posture this course asks for — neither 'it will never work' nor 'it is nearly here', both of which are claims made without the resource counts that would settle them (Module 13).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Shor — Scheme for reducing decoherence in quantum memory	§§1 and 2 — the first quantum code, and the phase-flip problem solved.
Fowler et al. — Surface codes: towards practical large-scale quantum computation (free)	§§2 to 4 — the standard reference, with the overhead numbers and the decoding problem.
Preskill — Fault-tolerant quantum computation (free)	§3's threshold theorem , explained rather than only proved.
Google Quantum AI — below-threshold surface code results (free)	§§3 and 4 experimentally — a logical qubit outperforming its physical parts, which is the milestone to understand.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Show that I, X, Y, Z span the single-qubit operator space.**
- 2 **Explain discretisation of errors** in your own words.
- 3 **Say why phase flips have no classical analogue.**
- 4 **Work the three-qubit bit-flip code** by hand, with both parities.

- 5 **Confirm the parities reveal nothing** about the logical state.
- 6 **Simulate the three-qubit code** with an injected error and correct it.
- 7 **Try to correct a phase flip with it**, and see it fail.
- 8 **State the threshold theorem** and what it established.
- 9 **Compute the physical qubit count** for 1000 logical qubits at three overhead assumptions.
- 10 **Read one resource estimate** and extract its assumed physical error rate.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why do continuous errors not prevent correction?
- 2 Which errors must be corrected, and why does that suffice?
- 3 What is new relative to classical codes?
- 4 How is a syndrome measured without disturbing the data?
- 5 Why is the surface code the hardware favourite?
- 6 State the threshold theorem and what it established.
- 7 What is the approximate threshold, and where is hardware?
- 8 Give the overhead, and the two things that dominate it.
- 9 Why does T-count matter so much?
- 10 What should you read first in a resource estimate?